ALGEBRA REVIEW

BLAKE FARMAN

Lafayette College

Name: Solutions

Know how to do these problems without the aid of a book or notes.

1. Find all real solutions to each equation.

   (a) $x^2 - 8x + 12 = 0$

   $0 = x^2 - 8x + 12$
   $= (x-6)(x-2)$
   $\iff \boxed{x = 6 \text{ or } x = 2}$

   (b) $2x^2 - 9x = 5$

   $\iff 0 = 2x^2 - 9x - 5$
   $= (2x+1)(x-5)$
   $\iff \boxed{x = -1/2 \text{ or } x = 5}$

   (c) $x^2 - 1 = 0$

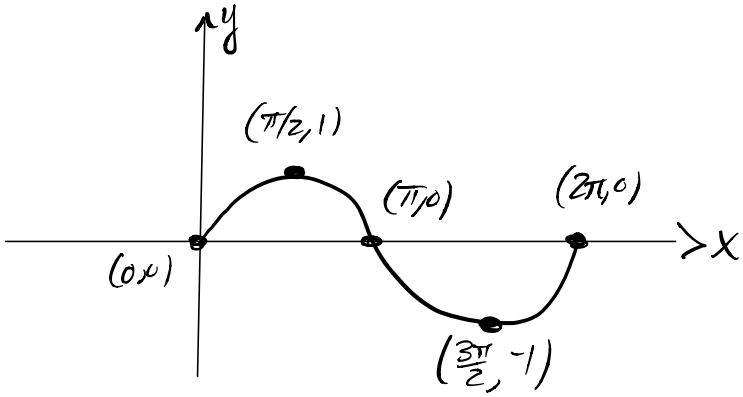
   $0 = x^2 - 1 = (x+1)(x-1)$
   $\iff \boxed{x = 1 \text{ or } x = -1}$

   (d) $x^2 = 2$

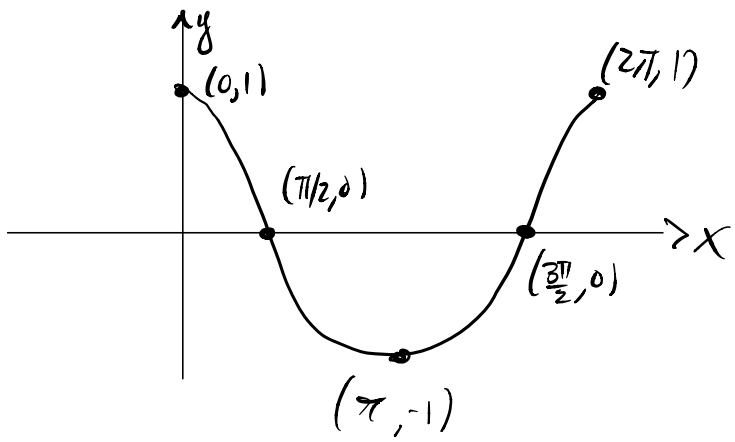
   $\boxed{x = \sqrt{2} \text{ or } x = -\sqrt{2}}$

2. Sketch a graph of the following:

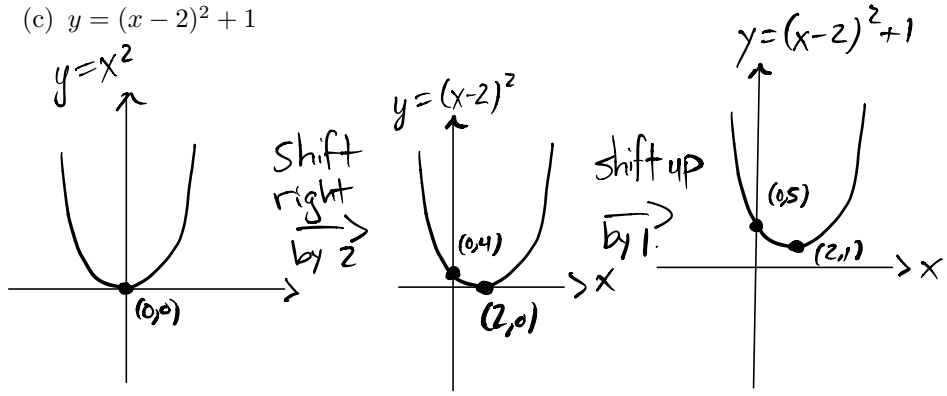
(a) $y = \sin(x)$



(b) $y = \cos(x)$

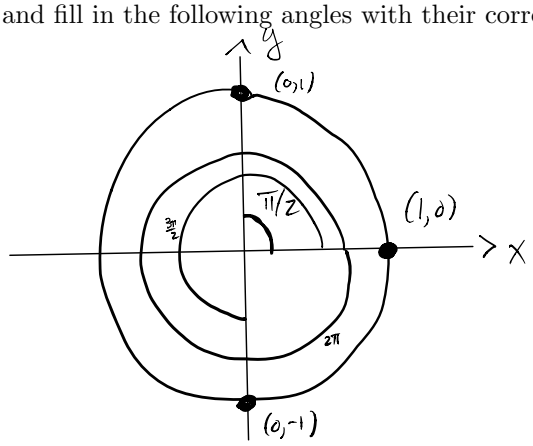


(c) $y = (x-2)^2 + 1$

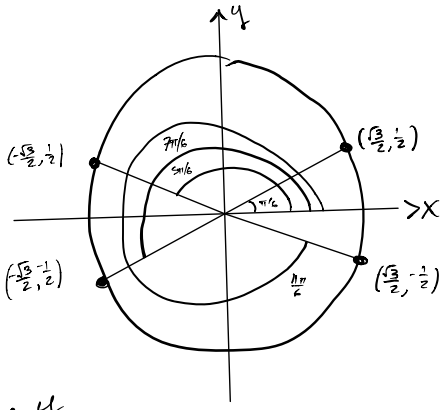


3. Draw the unit circle and fill in the following angles with their corresponding coordinates:

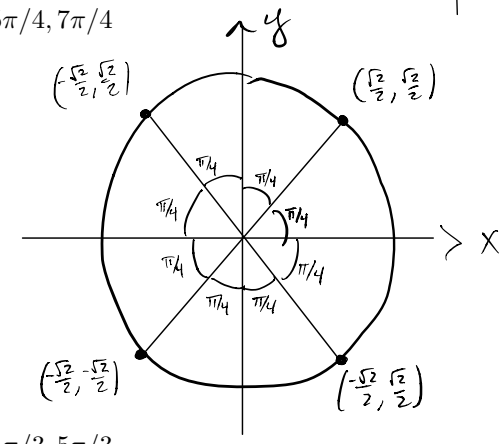
(a) $\pi/2, \pi, 3\pi/2, 2\pi$



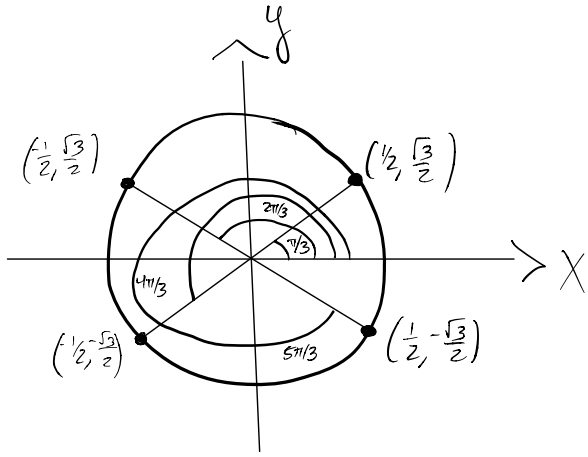
(b) $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$



(c) $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$



(d) $\pi/3, 2\pi/3, 4\pi/3, 5\pi/3$



4. Simplify the following expressions:

(a) $x^2(4(x-2)^3) + 2x(x-4)^4$

$$4x^2(x-2)^3 = 4x^2\left(x^3 + 3(x^2)(-2) + 3(x)(-2)^2 + (-2)^3\right) = 4x^2(x^3 - 6x^2 + 12x - 8) = 4x^5 - 24x^4 + 48x^3 - 32x^2$$

$$+\ 2x(x-4)^4 = 2x\left(x^4 + 4(x^3)(-4) + 6(x^2)(-4)^2 + 4(x)(-4)^3 + (-4)^4\right) = 2x(x^4 - 16x^3 + 96x^2 - 256x + 256) = 2x^5 - 32x^4 + 192x^3 - 512x^2 + 512x$$

$$\boxed{6x^5 - 56x^4 + 240x^3 - 544x^2 + 512x}$$

(b) $\dfrac{(x^2+3)^2(6) - 6x(2)(x^2+3)(2x)}{(x^2+3)^4}$

$$= \frac{6(x^2+3)\left[x^2+3 - 4x^2\right]}{(x^2+3)^4} = \frac{6(-3x^2+3)}{(x^2+3)^3} = \boxed{\frac{-18x^2+18}{(x^2+3)^3}}$$

(c) $\dfrac{\frac{1}{x^2} - \frac{1}{9}}{x-3}$

$$= \frac{1}{x-3}\left(\frac{9-x^2}{9x^2}\right) = \frac{(3-x)(3+x)}{(x-3)(9x^2)} = \frac{-(x-3)(3+x)}{(x-3)(9x^2)} = \boxed{\frac{-3-x}{9x^2}}$$

(d) $\dfrac{\sqrt{25+x^2} - x(1/2)(25+x^2)^{-1/2}(2x)}{25+x^2}$

$$= \frac{1}{25+x^2}\left(\sqrt{25+x^2} - \frac{x^2}{\sqrt{25+x^2}}\right) = \frac{1}{25+x^2}\left(\frac{25+x^2}{\sqrt{25+x^2}} - \frac{x^2}{\sqrt{25+x^2}}\right)$$

$$= \frac{1}{25+x^2}\left(\frac{25}{\sqrt{25+x^2}}\right) = \boxed{\frac{25}{(25+x^2)^{3/2}}}$$

5. Solve the following inequalities:

(a) $\frac{x}{2} - 1 < 3x + 9$

$\Rightarrow -1-9 = -10 < 3x - \frac{x}{2} = \frac{6x}{2} - \frac{x}{2} = \frac{5x}{2}$

$\Rightarrow -10(2/5) = \boxed{-4 < x}$

(b) $x + 3 < 2x + 8 < 3x + 10$

$x+3 < 2x+8$

$\Rightarrow 3-8 = -5 < 2x - x = x$

So $x+3 < 2x+8$ is true when $-5 < x$

$2x+8 < 3x+10$

$\Rightarrow 8-10 = -2 < 3x - 2x = x$

So $2x+8 < 3x+10$ is true when $-2 < x$

Therefore $x+3 < 2x+8 < 3x+10$ is true when $-5 < x$ and $-2 < x$, or, simply $\boxed{-2 < x}$

(c) $|2x - 5| \leq 11$

$\Leftrightarrow -11 \leq 2x - 5 \leq 11$

$\Leftrightarrow -6 \leq 2x \leq 16$

$\Leftrightarrow \boxed{-3 \leq x \leq 8}$